Bhanoth Sai chaitanya
2023102
CSE - B1

Q1) man → to see manual
$ man ls
$ man cd
ls → list file and directory
$ ls
$ lsl
cd → change working directory
$ cd
$ cd : /Directory name
Pwd → Print working directory
$ Pwd
mkdir → make directory
$ mkdir Directory name

mkdir → remove directory
$ rmdir : /Directory name
Cat → display file Content
$ Cat file.txt
less → Pages program used to view the Content
$ less
head → display new lines of a file
$ head -V
$ head -n 10

tail → display few lines from last of a file
$ tail -n 2

Bhavath Sai Chaitanya
2023102
CSE - B1

CP → Copy the file Src → dist
$ man cp
$ cp
$ cp -L

mv → move file
$ mv txt Directory name

rm → remove file or directory
$ rm t.txt

touch → create an empty file
$ touch text.txt

echo → out the string that is passed
$ echo "Hello World"
$ echo text.txt

Ping → $ Ping google.com

Chmod → changes access mode of a file or directory
$ man chmod
$ chmod -w hello.txt

Chown → change ownership
$ chown root hello.txt

Exit → terminates the current program
$ exit

Bhavath Sai Chaitanya
2023102
CSE - B1

Grep → Search pattern in a file
$ grep "hello"
$ grep -i 'hello'
$ grep -e [x-z]00

diff → Show difference between two files
$ diff hello.txt hello2.txt

PS → Show working process
$ PS
$ PS -l
$ PS -P4000

top → Table of process

kill → kill process

Sudo → run as admin

Shutdown → Shut downs the system

wc → Word Count
$ wc hello.txt

Bhanoth Sai Chaitanya
2023102
CSE - B1

Q2) (i) CPU info
Cat /proc/cpuinfo
Cat /proc/ CPU info | grep "model name"
Cat /proc/ CPU info | grep " CPU Core"
Cat /proc/ CPU info | grep " CPU Cores"

(ii) memory info
Vim Std
free -m
Cat /proc/ meminfo

Q3) (i) cd

```
#include<stdio.h>
#include<sys/stat.h>
#include<sys/types.h>
#include<sys/stat.h>

int main(int argc, char *argv[]){
    if (argc!=2){
        Printf("usage: %s <dir name>\n",
                argv[0]);
        return 1; }
    int status = chdir(argv[1]);
    if (status!=-1) Printf("%s directory changed\n",argv[1]);
    else Printf("%s directory not changed");
    char * cwd = getcwd(NULL,0);
```

Bhanoth Sai Chaitanya
2023102
CSE - B1

```
    if (cwd==NULL){
        Printf("getcwd failed");
        return 1;
    }
    else{
        Printf("%s directory not changed",argv[1]);
    } return 0; }
```

(ii) ls

```
#include<stdio.h>
#include<string.h>
#include<sys/types.h>
#include<dirent.h>
int main() { int argc, char *argv[]{
    DIR *dir;
    struct dirent *entry;
    bool list=false;
    char *pwd = getcwd(dir->Path name)
    if (argc>2 || argv[1][0]=='-'){
        int len = strlen(argv[argc-1]);
        if (len>=2){
            if (argv[1][1]=='a') show all true;
        else if (argv[1][1]=='l')list=true;
        else{ Printf("Invalid argv[0]");
            return 02;
        }
        if (len>=3){ if(argv[1][2]=='a')show-all
                                        true
        else if (argv[1][2]=='l') list = true;
```

Bhanoth Sai Chaitanya
2023102
CSE B1

```
else {
        Printf (fmt, argv[0]);
        return 2;
    }
}
if (argc == 1 || argv[argc-1][0] == '-')
        dir = opendir(".");
        dir = opendir(argv[argc-1]);
        return 3;
}
while (entry = readdir(dir) != NULL){
    if (!show all && entry dir name[0] == '.')
        continue;
    Printf ("%s", entry->d_name)
    if (list) printf("\n");
}
Printf ("\n")
Close dir (dir)
return 0;
}
```

(iii) mkdir

```
#include <stdio.h>
#include <sys/stat.h>
#include <sys/types.h>
int main (int argc, char **argv){
    if (argc < 2){
        Printf ("usage: mkdir <dir name>");
        return 1; }
```

Bhanoth Sai chaitanya
2023102
CSE B1

```
int status = mkdir(argv[i]) S_IRWX-U);
if (status == -1){
    Printf ("%s directory created \n", argv[i]);
}
```

(iv) Grep

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
    void Grep (Const char *filename, Const char* keyword){
    FILE *file = fopen (filename, "r");
    if (file == NULL){
        Printf ("error opening file");
        Exit (EXIT_FAILURE);
    }
    Char line[MAX-LINE];
    int line number = 0;
    While (fgets(line, MAX-LINE, file){
        line number ++;
    if (str str (line, keyword))
            Printf ("line %d : %s line number);
    fclose (file); }
int main (int argc, char ** argv){
    if (argc != -1){
        print f ("usage %
            (filename<keyword)")
    Grep (argv[1], argv[2]);
    return 0;
}
```

Bhanoth Sai Chaitanya
2023102
CSE - B1

Q4) The above c program demonstrates the use of Command line Arguments. This program calculates the sum of integers passed as command line arguments

```
# include <stdio.h>
# include <stdlib.h>
int main (int argc, char * argv [] ){
if (argc < 2){
    Printf("Usage: %s <number1> <number2> ...
        <number n>\n", argv[0]);
    } return 1;
    int sum = 0;
    for(int i=1; i<argc; i++){
        sum += atoi(argv[i]);
    }
Printf("The sum of the numbers is %d\n", sum);
    return 0;
}
```

Q5)

Bhanoth Sai Chaitanya
2023102
CSE B1

Q5)

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <limits.h>

void Swap(int *a, int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}
void SelectionSort(int arr[], int n){
start = clock(); for (int i=0; i<n-1; i++){
    int min = i;
    for (int j=i+1; j<n; j++){
        if (arr[j] < min) min = j;
    }
    if (min != i) Swap(&arr[min], &arr[i]);
}
end = clock();
double time = end - start;
Printf("Time of execution of selection Sort: %f\n",
                time);
}
Void BubbleSort(int arr, int n){
    start = clock();
    for (int i=0; i<n; i++){
```

Bhanoth Sai Chaitanya

2023102

CSE: B1.

```
for (int j=0 ; j<n-i-1 ; j++){
  if (arr[j] > arr[j+1]){
    Swap (&arr[j], &arr[j+1]);
  }
}
  end= clock();
double time = end-time;
Printf ("Execution time for insertion Sort : %.if", time);
}
Void Insertion Sort (int arr[], int n){
    start = clock();
    for (int i=1; i<n; i++) {
        int j = i-1;
      while (j>=0 && arr[j] > arr[i]){
        Arr[j+1] = arr[j];
        } for j--;

  arr[j+1] = arr[i];
}
  end = clock();
  double time = end-start;
Printf ("Execution Time for insertion Sort : %.if", time);
}
```